

Solving Trigonometric Equations on Restricted Intervals

Answer Key

Precalculus Trigonometric Identities and Equations

Quick Answers

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| 1. $\frac{\pi}{6}, \frac{5\pi}{6}$
2. $\frac{5\pi}{6}, \frac{7\pi}{6}$
3. $-\frac{\pi}{4}, \frac{3\pi}{4}$
4. $1 - 2\sin^2(x)$
5. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$ | 6. $0, \frac{2\pi}{3}, \frac{4\pi}{3}$
7. 13
8. 10, 20
9. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$
10. 90, 210, 330 |
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1. Solve $2\sin x - 1 = 0$ on $[0, 2\pi)$.

Solution. Isolate the trig function: $\sin x = \frac{1}{2}$. The reference angle is $\frac{\pi}{6}$, and sine is positive in quadrants I and II, so the two solutions in one revolution are $\frac{\pi}{6}$ and $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. $x = \frac{\pi}{6}, \frac{5\pi}{6}$

Count check: the graph crosses $y = \frac{1}{2}$ twice on $[0, 2\pi)$, matching the two answers.

2. Solve $2\cos x + \sqrt{3} = 0$ on $[0, 2\pi)$.

Solution. $\cos x = -\frac{\sqrt{3}}{2}$. The reference angle is $\frac{\pi}{6}$; cosine is negative in quadrants II and III, giving $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$ and $\pi + \frac{\pi}{6} = \frac{7\pi}{6}$. $x = \frac{5\pi}{6}, \frac{7\pi}{6}$

3. Solve $\tan x + 1 = 0$ on $[-\pi, \pi)$.

Solution. $\tan x = -1$, reference angle $\frac{\pi}{4}$. Tangent has period π , so its solutions are spaced π apart: $x = -\frac{\pi}{4} + k\pi$. The members of that family lying in $[-\pi, \pi)$ are $-\frac{\pi}{4}$ (from $k = 0$) and $-\frac{\pi}{4} + \pi = \frac{3\pi}{4}$ (from $k = 1$). $x = -\frac{\pi}{4}, \frac{3\pi}{4}$ *Common error:* using period 2π and reporting only one solution, or listing $\frac{7\pi}{4}$, which lies outside this interval.

4. Verify $\cos 2x = 1 - 2\sin^2 x$.

Solution. Start from the difference form of the double-angle identity and replace $\cos^2 x$ using the Pythagorean identity:

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x \\ &= (1 - \sin^2 x) - \sin^2 x \\ &= 1 - 2\sin^2 x\end{aligned}$$

$\cos 2x = 1 - 2\sin^2 x$ This is the form to reach for whenever an equation mixes $\cos 2x$ with $\sin x$: it makes every term a function of the same angle.

5. Solve $\cos 2x + 3 \sin x = 2$ on $[0, 2\pi)$.

Solution. Substitute the identity from problem 4 so that every term is a function of $\sin x$:

$$\begin{aligned}(1 - 2 \sin^2 x) + 3 \sin x &= 2 \\ -2 \sin^2 x + 3 \sin x - 1 &= 0 \\ 2 \sin^2 x - 3 \sin x + 1 &= 0 \\ (2 \sin x - 1)(\sin x - 1) &= 0\end{aligned}$$

So $\sin x = \frac{1}{2}$, giving $x = \frac{\pi}{6}$ and $\frac{5\pi}{6}$; or $\sin x = 1$, giving $x = \frac{\pi}{2}$. $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$

6. Solve $2 \cos^2 x - \cos x - 1 = 0$ on $[0, 2\pi)$.

Solution. Treat $\cos x$ as the variable and factor:

$$(2 \cos x + 1)(\cos x - 1) = 0$$

$\cos x = -\frac{1}{2}$ gives the quadrant II and III angles $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$; $\cos x = 1$ gives $x = 0$ (and 2π , which is excluded because the interval is half-open). $x = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$ *Grading note:* an answer that includes 2π has misread the interval; an answer missing $x = 0$ has divided by $(\cos x - 1)$ instead of factoring.

7. Ferris wheel height at $t = 10$ seconds, $h(t) = 9 - 8 \cos(\pi t/15)$.

Solution. $\frac{\pi(10)}{15} = \frac{2\pi}{3}$, and $\cos \frac{2\pi}{3} = -\frac{1}{2}$, so $h(10) = 9 - 8(-\frac{1}{2}) = 9 + 4$. 13 m *Sanity check:* the model has centre height 9 m and radius 8 m, so every value must lie between 1 m and 17 m; 13 m does.

8. Find every $t \in [0, 30)$ seconds with $h(t) = 13$ metres.

Solution. Set the model equal to the target and isolate the cosine:

$$\begin{aligned}9 - 8 \cos\left(\frac{\pi t}{15}\right) &= 13 \\ \cos\left(\frac{\pi t}{15}\right) &= -\frac{1}{2}\end{aligned}$$

Let $u = \frac{\pi t}{15}$. As t runs over $[0, 30)$, u runs over $[0, 2\pi)$ — exactly one revolution — so $u = \frac{2\pi}{3}$ or $u = \frac{4\pi}{3}$. Converting back, $t = \frac{15u}{\pi}$ gives $t = 10$ and $t = 20$. $t = 10 \text{ s}, 20 \text{ s}$ The two times are symmetric about $t = 15$ s, the top of the wheel — the rider passes 13 m once going up and once coming down.

9. Solve $\sin 2x = \cos x$ on $[0, 2\pi)$.

Solution. Rewrite the double angle, move everything to one side, and factor — never divide by $\cos x$, which would delete the solutions where $\cos x = 0$:

$$\begin{aligned} 2 \sin x \cos x - \cos x &= 0 \\ \cos x (2 \sin x - 1) &= 0 \end{aligned}$$

$\cos x = 0$ gives $x = \frac{\pi}{2}$ and $\frac{3\pi}{2}$; $\sin x = \frac{1}{2}$ gives $x = \frac{\pi}{6}$ and $\frac{5\pi}{6}$. $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ *Grading note:* exactly the two answers a student loses by dividing through by $\cos x$ are $\frac{\pi}{2}$ and $\frac{3\pi}{2}$.

10. Solve $2 \sin^2 \theta - \sin \theta - 1 = 0$ on $[0^\circ, 360^\circ)$.

Solution. Factor the quadratic in $\sin \theta$:

$$(2 \sin \theta + 1)(\sin \theta - 1) = 0$$

$\sin \theta = 1$ gives $\theta = 90^\circ$. $\sin \theta = -\frac{1}{2}$ has reference angle 30° and is negative in quadrants III and IV, giving $180^\circ + 30^\circ = 210^\circ$ and $360^\circ - 30^\circ = 330^\circ$. $\theta = 90^\circ, 210^\circ, 330^\circ$ *Grading note:* $\sin \theta = 1$ contributes only one solution per revolution because 1 is the maximum of sine — a student who reports two angles there has applied the quadrant rule without checking the extreme value.